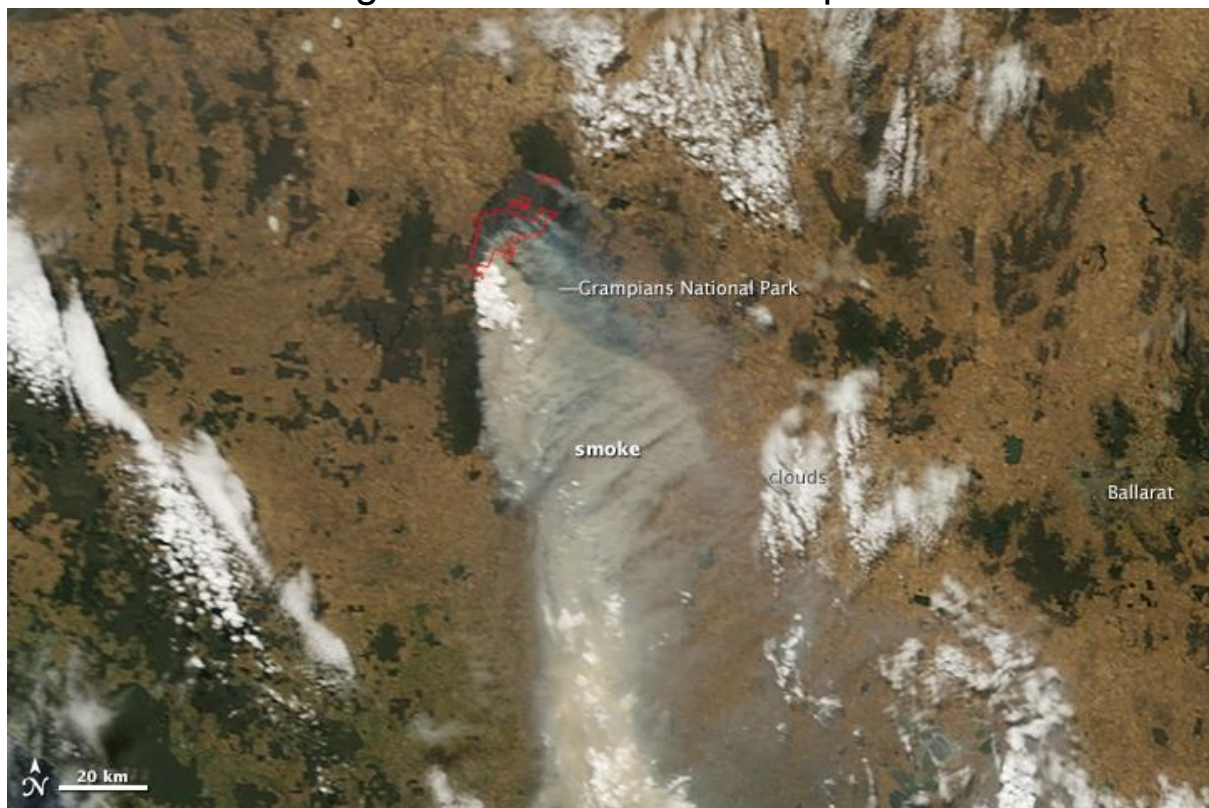


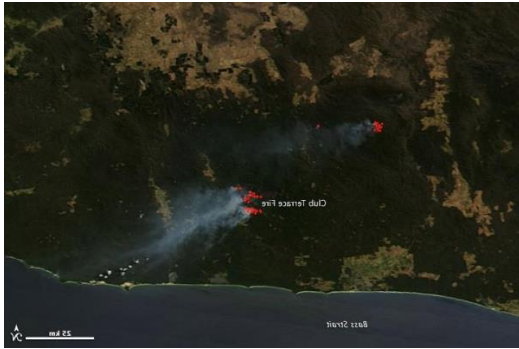
# Bushfire Analysis, Grampians National Park, Victoria, Australia

## Introduction:

By January 2014, an intense blaze that was burning 43,000 hectares (106,000 acres) in and around Grampians National Park in Victoria, Australia, had spread. In Mount Difficult Range, the fire was burning close west of Hallis Gap.



This natural-color satellite image was acquired by the MODIS Satellite on January 17, 2014. Actively burning areas, detected by MODIS's thermal bands are outlined in red.



This image was taken on January 28, 2014, while two sizable bushfires were burning in Victoria, Australia, by NASA's Aqua satellite using the Moderate Resolution Imaging Spectroradiometer (MODIS). Red markers indicate the fires. The Victorian government claims that 5,567 hectares had been burned in the Club Terrace fire, which appears to be the largest of the two fires.

In Victoria, February is often the peak of the fire season. The state has dispatched more firemen in anticipation of an intense heatwave to be ready for any potential fires.

### **Managing Bushfires with GIS and AI:**

Australian authorities and fire departments can now prepare for, respond to, and recover from bushfires with the use of GIS technology, which has become a potent instrument. Around the world, wildfires destroying large areas of the United States and Canada are managed and contained by technology.

### **Data and methods:**

Using satellite photos to detect active fires is a binary classification challenge. The satellite imagery has labels indicating whether a pixel is "non-fire" or "fire." Another name for the "fire" pixels is hotspots. We will use the satellite photography, including its classification system for identifying hotspots.